Theorem 33. Imp **2017**

Let X has mean 'p' and variance pq and

$$S_n = x_1 + x_2 + \cdots + x_n \sim B(n, p)$$

then show that:

i) show that $\frac{S_n}{n}$ is unbiased estimator of p

ii) show that $1 - \frac{S_n}{n}$ is unbiased estimator of $1-p$

iii) show that $\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)$ is not unbiased estimator of pq

Find Bias and show that $\frac{S_n}{n}\left[1 - \frac{S_n}{n}\right]$ is asymptotically unbiased

iv) show that $\frac{S_n}{n}$ is consistent in mean square as an estimator of p

v) prove that $\frac{S_n}{n}\left[1 - \frac{S_n}{n}\right]$ is a consistent estimator of pq.

Proof

(i)

$$S_n = x_1 + x_2 + \cdots + x_n$$

Apply Expectation

$$E(S_n) = E\left[x_1 + x_2 + \cdots + x_n\right]$$

$$= E(x_1) + E(x_2) + \cdots + E(x_n)$$

$$E(S_n) = p + p + \cdots + p \text{ (n times)}$$

$$E(S_n) = np$$

$\div$ by n both sides

$$E\left(\frac{S_n}{n}\right) = \frac{\cancel{n}p}{\cancel{n}}$$

$$E\left(\frac{S_n}{n}\right) = p$$

This show that $\frac{S_n}{n}$ is unbiased estimator of p

ii

$$E\left[1 - \frac{S_n}{n}\right] = 1 - E\left(\frac{S_n}{n}\right)$$

$$= 1 - p = q$$

This shows that $1 - \frac{S_n}{n}$ is unbiased estimator of $1-p$

iii

$$E\left[\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)\right] = E\left[\frac{S_n}{n} - \frac{S_n^2}{n^2}\right]$$

$$= E\left(\frac{S_n}{n}\right) - E\left(\frac{S_n^2}{n^2}\right) \quad (1)$$

$$\because V(S_n) = E(S_n^2) - \left[E(S_n)\right]^2$$

$$E(S_n^2) = V(S_n) + \left[E(S_n)\right]^2$$

$$= npq + n^2p^2$$

$\div$ by n^2 b.s

$$E\left(\frac{S_n^2}{n^2}\right) = \frac{npq + n^2p^2}{n^2}$$

Put in eq (i)

$$E\left[\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)\right] = p - \frac{npq + n^2p^2}{n^2}$$

$$= p - \frac{\cancel{n}(pq + np^2)}{n^{\cancel{2}}}$$

$$= p - \frac{pq + np^2}{n}$$

$$= \frac{np - pq - np^2}{n}$$

$$= \frac{1}{n}\left[np - np^2 - pq\right]$$

$$= \frac{1}{n}\left\{np(1-p) - pq\right\}$$

$$= \frac{1}{n}(npq - pq)$$

$$= \frac{pq}{n}(n-1)$$

$$E\left[\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)\right] = \frac{(n-1)}{n}pq \quad \text{Biased}$$

Now $\quad \text{Bias} = E(\hat{\theta}) - \theta$

$$= \frac{(n-1)}{n}pq - pq$$

$\boxed{\text{Bias of } \frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)}$

$$= pq\left[\frac{n-1}{n} - 1\right]$$

$$= pq\left[\frac{\cancel{n} - 1 - \cancel{n}}{n}\right]$$

$$\text{Bias} = \frac{-pq}{n}$$

Apply Limit

$$\lim_{n\to\infty} \text{Bias} = \frac{-pq}{\infty}$$

$$= 0$$

So, $\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)$ is Assymptotically unbiased.

iv

$$MSE = E\left[\overset{*}{\theta} - \theta\right]^2$$

$$= E\left[\frac{S_n}{n} - p\right]^2$$

$$= V\left(\frac{S_n}{n}\right)$$

$$= \frac{1}{n^2}V(S_n)$$

$$= \frac{1}{n^2}npq$$

$$V(x) = E\left[x - E(x)\right]^2$$

$$= E\left[\frac{S_n}{n} - E\left(\frac{S_n}{n}\right)\right]^2$$

$$V\left(\frac{S_n}{n}\right) = E\left[\frac{S_n}{n} - p\right]^2$$

$$MSE = \frac{pq}{n}$$

Apply Limit

$$\lim_{n\to\infty} MSE = \frac{pq}{\infty} = 0 \quad \text{(Consistent)}$$

v

$$E\left[\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)\right] = \frac{n-1}{n}pq$$

$$= \left(1 - \frac{1}{n}\right)pq$$

$$= pq - \frac{pq}{n}$$

Apply Limit

$$\lim_{n\to\infty} E\left[\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)\right] = pq - \frac{pq}{\infty}$$

$$= pq - 0$$

$$= pq$$

which shows that $\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)$ is a ~~Consis~~ Consistent estimator of pq.

2017 q#4

Theorem 44: Imp

Let $X_1, \ldots, X_n$ be an ~~idempotent~~ independent random sample from $f(x) = \frac{1}{\theta}$ $\quad 0 < x < \theta$.
Show that $Y_n = \max(X_i)$ is a sufficient statistic for θ but $Y_1 = \min(X_i)$ is not a sufficient statistic for θ.

Proof.

$$f(x) = \frac{1}{\theta}$$

$$L(x) = f(x_1, \ldots, x_n)$$

$$= \prod_{i=1}^{n} f(x_i)$$

$$L(x) = \frac{1}{\theta} \cdot \frac{1}{\theta} \cdots \frac{1}{\theta} = \frac{1}{\theta^n}$$

(i)

$$g(y_n) = n\left[F(y_n)\right]^{n-1} f(y_n)$$

$$F(y_n) = \int_0^{y_n} \frac{1}{\theta}\, dy_n$$

$$= \frac{1}{\theta}\left|y_n\right|_0^{y_n}$$

$$= \frac{y_n}{\theta}$$

$$g(y_n) = n\left[\frac{y_n}{\theta}\right]^{n-1} \cdot \frac{1}{\theta}$$

$$= \frac{n}{\theta^n} y_n^{n-1}$$

The conditional distribution is:

$$p(x_1, \ldots, x_n / y_n) = \frac{\text{Joint distribution}}{\text{Marginal distribution}}$$

$$= \frac{1/\theta^n}{\frac{n}{\theta^n} y_n^{n-1}}$$

$$= \frac{1}{n\, y_n^{n-1}}$$

It is free from parameter.
Hence $Y_n = \max(x_i)$ is sufficient statistic for θ.

(ii)

$$f(x) = \frac{1}{\theta}$$

$$L(x) = f(x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i)$$

$$= \frac{1}{\theta} \cdot \frac{1}{\theta} \cdots \frac{1}{\theta} = \frac{1}{\theta^n}$$

$$g(y_1) = n\left[1 - F(y_1)\right]^{n-1} f(y_1)$$

$$F(y_1) = \int_0^{y_1} \frac{1}{\theta}\, dy_1$$

$$= \frac{1}{\theta}\left|y_n\right|_0^{y_1}$$

$$= \frac{y_1}{\theta}$$

$$g(y_1) = n\left[1 - \frac{y_1}{\theta}\right]^{n-1} \frac{1}{\theta}$$

$$= n\left[\frac{\theta - y_1}{\theta}\right]^{n-1} \frac{1}{\theta}$$

$$= \frac{n}{\theta^{n-1}\,\theta}\left[\theta - y_1\right]^{n-1}$$

$$g(y_1) = \frac{n}{\theta^n}(\theta - y_1)^{n-1}$$

The conditional distribution is:

$$p(x_1, \ldots, x_n / y_1) = \frac{1/\theta^n}{n/\theta^n\,(\theta - y_1)^{n-1}}$$

$$= \frac{1}{n(\theta - y_1)^{n-1}}$$

It is not free from θ, Hence $Y_1 = \min(X_i)$ is not a sufficient statistic of θ.

Proved

Ex. Given $f(x;\theta) = 1/\theta$, $0 < x < \theta$, $\theta > 0$

Compute the reciprocal of $nE\left[\frac{\partial \ln f(x;\theta)}{\partial\theta}\right]^2$ and compare this with the variance of $\left(\frac{n+1}{n}\right)Y_n$, where Y_n is the largest item of the random sample of size 'n' of this distribution.

Show that the least attainable variance may be less than the CRLB.

2017 q#5

Sol: $f(x) = 1/\theta = \theta^{-1}$

$\ln f(x) = -1 \ln\theta$

$\frac{\partial \ln f(x)}{\partial\theta} = -1/\theta$

$\left[\frac{\partial \ln f(x)}{\partial\theta}\right]^2 = 1/\theta^2$

$nE\left[\frac{\partial \ln f(x)}{\partial\theta}\right]^2 = nE(1/\theta^2) = n/\theta^2$

So the reciprocal of $nE\left[\frac{\partial \ln f(x)}{\partial\theta}\right]^2 = \theta^2/n$ —(i)

Date: __/__/20__

The dist^n of n^{th} order statistic is Y_n and its f^n be

as $g(y_n) = n\,[F(y_n)]^{n-1} f(y_n)$

$F(y) = \int_0^y 1/\theta\, dy = y/\theta$

$g(y_n) = n\,(y/\theta)^{n-1}\, 1/\theta = \frac{n}{\theta^n} y^{n-1} \qquad 0 \le y \le \theta$

$E(y_n^r) = \int_0^\theta y_n^r\, g(y_n)\, dy_n = \int_0^\theta y^r \frac{n}{\theta^n} y^{n-1} dy$

$= \frac{n}{\theta^n}\int_0^\theta y^{n+r-1} dy = \frac{n}{\theta^n}\left[\frac{y^{n+r}}{n+r}\Big|_0^\theta\right]$

$= \frac{n}{\theta^n(n+r)}\left[\theta^{r+n} - 0\right] = \frac{n\theta^r}{n+r}$ —(2)

put $r=1$ and $r=2$ in eq (2)

$E(y_n) = \frac{n\theta}{n+1}$, $E(y_n^2) = \frac{n\theta^2}{n+2}$ —(3)

$E\left[\left(\frac{n+1}{n}\right)Y_n\right] = \frac{n+1}{n}E(Y_n) = \frac{n+1}{n}\cdot\frac{n\theta}{n+1} = \theta$

$\Rightarrow \left(\frac{n+1}{n}\right)Y_n$ is unbiased for θ.

$V\left(\left(\frac{n+1}{n}\right)Y_n\right) = \left(\frac{n+1}{n}\right)^2 V(Y_n) = \left(\frac{n+1}{n}\right)^2\left[E(Y_n^2) - \{E(Y_n)\}^2\right]$

$= \left(\frac{n+1}{n}\right)^2\left[\frac{n\theta^2}{n+2} - \frac{n^2\theta^2}{(n+1)^2}\right]$

$= \frac{(n+1)^2}{n^2}\left[\frac{n\theta^2(n+1)^2 - n^2\theta^2(n+2)}{(n+2)(n+1)^2}\right]^{**} = \frac{\theta^2}{n(n+2)}$ —(5)

comparing with (i)

Date: __/__

$\frac{\theta^2}{n(n+2)} \le \theta^2/n$, that is the

var$\left(\left(\frac{n+1}{n}\right)Y_n\right)$ is less than CRLB.

$** \quad \frac{(n+1)^2}{n^2}\left[\frac{n\theta^2}{n+2} - \frac{n^2\theta^2}{(n+1)^2}\right]$

$= \frac{(n+1)^2\theta^2}{n^2}\left[\frac{n^2}{n(n+2)} - \frac{n^2}{(n+1)^2}\right]$

$= \frac{n^2(n+1)^2\theta^2}{n^2}\left[\frac{1}{n(n+2)} - \frac{1}{(n+1)^2}\right]$

$= \cancel{(n+1)^2}\theta^2\left[\frac{(n+1)^2 - n(n+2)}{n(n+2)\cancel{(n+1)^2}}\right]$

$= \frac{\theta^2}{n(n+2)}\left[n^2 + 1 + 2n - n^2 - 2n\right]$

$= \frac{\theta^2}{n(n+2)}$

Q. No. 6. State and prove Rao Blackwell theorem. pdf

Book pg (61)

2018 Q# 03

DATE: DAY:

Inference

let $X_1, \ldots, X_6$ is an identically distributed random Sample from a normal distribution with mean μ and variance σ^2. Determine 'c' such that $c[(X_1-X_2)^2+(X_3-X_4)^2+(X_5+X_6)^2]$ is an unbiased estimator of σ^2.

$$c \cdot E[(X_1-X_2)^2+(X_3-X_4)^2+(X_5-X_6)^2] = \sigma^2$$

$$c \cdot E[X_1^2+X_2^2-2X_1X_2+X_3^2+X_4^2-2X_3X_4+X_5^2+X_6^2 -2X_5X_6] = \sigma^2$$

$$E(X_1^2) = E(X_2^2) = E(X_3^2) = E(X_4^2) = E(X_5^2) = E(X_6^2) = \mu^2+\sigma^2$$

$$var(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = var(X) + [E(X)]^2$$

$$E(X^2) = \sigma^2 + \mu^2$$

$$E(XY) = E(X)E(Y)$$

$$E(X_1X_2) = E(X_1)E(X_2) = \mu \cdot \mu = \mu^2$$

$$c[\cancel{\mu^2}+\sigma^2+\cancel{\mu^2}+\sigma^2-\cancel{2\mu^2}+\cancel{\mu^2}+\sigma^2+\cancel{\mu^2}+\sigma^2-\cancel{2\mu^2} +\cancel{\mu^2}+\sigma^2+\cancel{\mu^2}+\sigma^2-\cancel{2\mu^2}]$$

$$6c\sigma^2 = \sigma^2$$

$$\boxed{c = \frac{1}{6}}$$

q#5

Theorem. 41

If E_1 and E_2 are the efficiencies of the two unbiased estimators t_1 and t_2 of $g(\theta)$ and ρ (rho) is the correlation coefficient between t_1 and t_2 then.

$$\sqrt{E_1E_2} - \sqrt{(1-E_1)(1-E_2)} \le \rho \le \sqrt{E_1E_2} + \sqrt{(1-E_1)(1-E_2)}$$

and if E_1 or $E_2 = 1$ then show that

$\rho = \sqrt{E}$

proof

Suppose t is the minimum variance unbiased estimator (MVUE) of $g(\theta)$ with variance 'v' and t_1 ~~and~~ and t_2

are any two unbiased estimators of $g(\theta)$ with finite variance. Then consider a new estimator $t_3 = at_1 + (1-a)t_2$ this is also an estimator of $g(\theta)$ with variance:

$$v(t_3) = a^2 v(t_1) + (1-a)^2 v(t_2) + 2a(1-a)\,Cov(t_1, t_2) \quad \text{—(i)}$$

Let $v(t_1) = k_1 v$, $v(t_2) = k_2 v$

where $k_1, k_2 > 1$

making this subsitution in eq (i)

$$v(t_3) = a^2k_1 v + (1-a)^2 k_2 v + 2a(1-a)Cov(t_1, t_2)$$

$\div$ by v b.s

$$\frac{v(t_3)}{v} = a^2k_1 + (1-a)^2k_2 + 2a(1-a)\frac{Cov(t_1, t_2)}{v} \quad \text{—(2)}$$

The correlation coefficient by t_1 and t_2 is:

$$\rho = \frac{Cov(t_1, t_2)}{\sqrt{v(t_1)\cdot v(t_2)}}$$

$$= \frac{Cov(t_1, t_2)}{\sqrt{(k_1 v)(k_2 v)}}$$

$$\rho = \frac{Cov(t_1, t_2)}{v(k_1 k_2)^{1/2}}$$

$$Cov(t_1, t_2) = \rho \cdot v \cdot (k_1 k_2)^{1/2}$$

eq (2) becomes

$$\frac{v(t_3)}{v} = a^2k_1 + (1-a)^2k_2 + 2a(1-a)\frac{\rho \cdot v \cdot (k_1k_2)^{1/2}}{v}$$

$$\frac{v(t_3)}{v} = a^2k_1 + (1-a)^2k_2 + 2a(1-a)\rho(k_1k_2)^{1/2}$$

But $v(t_3) \geq 1$. v is is the variance of MVUE

$\div$ by v b.s

$\frac{v(t_3)}{v} \geq 1$, therefore

$$a^2k_1 + (1-a)^2k_2 + 2a(1-a)\rho(k_1k_2)^{1/2} \geq 1 \quad \text{—(3)}$$

This may be re arranged as Quadratic ~~equation~~ inequalities in a

$$a^2k_1 + (1+a^2-2a)k_2 + (2a-2a^2)\rho(k_1k_2)^{1/2} \geq 1$$

$$a^2k_1 + k_2 + k_2a^2 - 2ak_2 + 2a\rho(k_1k_2)^{1/2} - 2a^2\rho(k_1k_2)^{1/2} \geq 1$$

$$a^2k_1 + a^2k_2 - 2a^2\rho(k_1k_2)^{1/2} - 2ak_2 + 2a\rho(k_1k_2)^{1/2} + k_2 - 1 \geq 0$$

$$a^2\{k_1 + k_2 - 2\rho(k_1k_2)^{1/2}\} + 2a\{\rho(k_1k_2)^{1/2} - k_2\} + (k_2 - 1) \geq 0$$

This quadratic equation as:

$ax^2 + bx + c = 0$

where

$a = k_1 + k_2 - 2\rho(k_1k_2)^{1/2}$

$b = 2\{\rho(k_1k_2)^{1/2} - k_2\}$

$c = k_2 - 1$

Since the roots of the quadratic equation are complex or equal so, the discriminant cannot be positive so the discriminant is negative.

$$b^2 - 4ac \le 0$$

$$[2\{\rho(k_1k_2)^{1/2} - k_2\}]^2 - 4\{k_1 + k_2 - 2\rho(k_1k_2)^{1/2}\}(k_2 - 1) \le 0$$

$$4\{\rho(k_1k_2)^{1/2} - k_2\}^2 - 4\{k_1 + k_2 - 2\rho(k_1k_2)^{1/2}\}(k_2 - 1) \le 0$$

$$\rho^2(k_1k_2) + \cancel{k_2^2} - 2\rho k_2(k_1k_2)^{1/2} - k_1k_2 - \cancel{k_2^2} + 2\rho k_2(k_1k_2)^{1/2} + k_1 + k_2 - 2\rho(k_1k_2)^{1/2} \le 0$$

$$\rho^2k_1k_2 - k_1k_2 + k_1 + k_2 - 2\rho(k_1k_2)^{1/2} \le 0$$

$$\rho^2k_1k_2 - 2\rho\sqrt{k_1k_2} \le k_1k_2 - k_1 - k_2$$

$$\rho^2k_1k_2 - 2\rho\sqrt{k_1k_2} + 1 \le k_1k_2 - k_1 - k_2 + 1$$

$$(\rho\sqrt{k_1k_2} - 1)^2 \le k_1(k_2 - 1) - 1(k_2 - 1)$$

$$(\rho\sqrt{k_1k_2} - 1)^2 \le (k_1 - 1)(k_2 - 1)$$

$$\rho\sqrt{k_1k_2} - 1 \le \pm\sqrt{(k_1-1)(k_2-1)}$$

$$\rho\sqrt{k_1k_2} \le 1 \pm \sqrt{(k_1-1)(k_2-1)}$$

$$\rho \le \frac{1}{\sqrt{k_1k_2}} \pm \frac{\sqrt{(k_1-1)(k_2-1)}}{\sqrt{k_1 k_2}}$$

$$\rho \le \frac{1}{\sqrt{k_1k_2}} \pm \sqrt{\frac{(k_1-1)(k_2-1)}{k_1k_2}}$$

$$\rho \le \frac{1}{\sqrt{k_1k_2}} \pm \sqrt{\frac{(k_1-1)}{k_1}\cdot\frac{k_2-1}{k_2}}$$

$$\rho \le \frac{1}{\sqrt{k_1k_2}} \pm \sqrt{\left(1-\frac{1}{k_1}\right)\left(1-\frac{1}{k_2}\right)}$$

As $E_1 = \frac{v}{v(t_1)} = \frac{v}{k_1v} = \frac{1}{k_1}$

$E_2 = \frac{v}{v(t_2)} = \frac{v}{kv} = \frac{1}{k_2}$

$$\rho \le \sqrt{E_1E_2} \pm \sqrt{(1-E_1)(1-E_2)}$$

$$\sqrt{E_1E_2} - \sqrt{(1-E_1)(1-E_2)} \le \rho \le \sqrt{E_1E_2} + \sqrt{(1-E_1)(1-E_2)}$$

Proved as

$E_2 = 1$

$$\sqrt{E_1} \le \rho \le \sqrt{E_1}$$

$\rho = \sqrt{E}$ Proved.

q#6

Uniqueness Minimum Variance Estimators

In many cases the MVB will not be attained, even when the conditions under which it is derived are satisfied. When this is so, there may be a minimum variance (MV) estimator for all values of θ.

Show that if a MV estimator exists, it is essentially unique, irrespective of whether any bound is attained.

Proof:

Let t_1 and t_2 be two minimum variance unbiased estimators of $T(\theta)$ each with variance 'V', we consider a new estimator: $t_3 = \frac{1}{2}(t_1 + t_2)$ which is also an unbiased estimator of $T(\theta)$ with variance,

$$V(t_3) = \frac{1}{4}\left[V(t_1) + V(t_2) + 2\,\mathrm{Cov}(t_1, t_2)\right] \quad \text{—(i)}$$

and

$$\mathrm{Cov}(t_1, t_2) = E\left[\{t_1 - E(t_1)\}\{t_2 - E(t_2)\}\right]$$

$$= E\left[\{t_1 - T(\theta)\}\{t_2 - T(\theta)\}\right]$$

$$= \int\int\cdots\int\left[\{t_1 - T(\theta)\}\{t_2 - T(\theta)\}\right] L \; dx_1 \ldots dx_n$$

$$[\mathrm{Cov}(t_1, t_2)]^2 \leq \left[\int\int\cdots\int \{t_1 - T(\theta)\}^2 L \, dx_1 \ldots dx_n\right] \left[\int\int\cdots\int \{t_2 - T(\theta)\}^2 L \, dx_1 \ldots dx_n\right]$$

$$[\mathrm{Cov}(t_1, t_2)]^2 \leq V(t_1)\,V(t_2)$$

$$\{\mathrm{Cov}(t_1, t_2)\}^2 \leq V.V$$

$$\mathrm{Cov}(t_1, t_2) \leq V \quad \text{—(2)}$$

From eq (i)

$$V(t_3) \leq \frac{1}{4}(V + V + 2V)$$

$V(t_3) \leq V$, which contradicts the assumption that t_1 and t_2 have minimum variance unless the equality holds in eq (2) and this can only be if

$$[t_1 - T(\theta)] = K(\theta)[t_2 - T(\theta)] \quad \text{—(3)}$$

i.e the two variables are proportional to each other. Multiply both sides by $[t_2 - T(\theta)]$

$$[t_1 - T(\theta)][t_2 - T(\theta)] = K(\theta)[t_2 - T(\theta)]^2$$

$$E\left[\{t_1 - T(\theta)\}\{t_2 - T(\theta)\}\right] = K(\theta)\,E[t_2 - T(\theta)]^2$$

$$\mathrm{Cov}(t_1, t_2) = K(\theta)\,V \quad \text{—(4)}$$

and thus equality sign hold in eq (2) i.e $\mathrm{Cov}(t_1, t_2) = V$, now equation (4) can be written as $V = K(\theta)V$

$K(\theta) = 1$, from equation (3)

$$[t_1 - T(\theta)] = K(\theta)[t_2 - T(\theta)]$$

$$[t_1 - T(\theta)] = 1\,[t_2 - T(\theta)]$$

$$t_1 - T(\theta) = t_2 - T(\theta)$$

$$t_1 = t_2$$

Thus the minimum variance estimator is unique.

2018

DATE: Q#4 DAY:

Given the p.d.f:

$f(X;\theta) = e^{-(x-\theta)}$ for $\theta < x < \infty$

for a random X_1, X_2, X_3 the joint p.d.f is:

$L(\theta; X_1, X_2, X_3) = f(X_1;\theta) \cdot f(X_2;\theta) \cdot f(X_3;\theta)$

$L(\theta; X_1, X_2, X_3) = e^{-(X_1-\theta)} \cdot e^{-(X_2-\theta)} \cdot e^{-(X_3-\theta)}$

$L\theta; X_1, X_2, X_3) = e^{-(X_1+X_2+X_3-3\theta)}$

Order the Sample values as $Y_1 < Y_2 < Y_3$ where $Y_3 = \max(X_i)$.

The joint p.d.f order statistics given as:

$f_{Y_1,Y_2,Y_3}(y_1, y_2, y_3;\theta) = 3! \cdot e^{-(y_1-\theta)} \cdot e^{-(y_2-\theta)} \cdot e^{-(y_3-\theta)}$

$f_{Y_1,Y_2,Y_3}(y_1, y_2, y_3;\theta) = 6 \cdot e^{-(y_1+y_2+y_3-3\theta)}$

$f_{Y_1,Y_2,Y_3}(y_1, y_2, y_3;\theta) = 6 \cdot e^{-3\theta} \cdot e^{-(y_1+y_2+y_3)}$

By using factorization theorem:

$T(X)$ is sufficient for θ.

The joint p.d.f can be factored into:

$L(\theta;X) = g(T(X),\theta) \cdot h(X)$

$T(X) = \max(X_i) = Y_3$

For $Y_3 = \max(X_i)$:

$f_{Y_3}(y_3;\theta) = \int_{-\infty}^{y_3}\int_{-\infty}^{y_3}\int_{-\infty}^{y_3} 6 \cdot e^{-3\theta} \cdot e^{-(y_1+y_2+y_3)}\, dy_1\, dy_2\, dy_3$

$$f_{Y_1,Y_2,Y_3}(y_1,y_2,y_3;\theta) = g(Y_3,\theta)\cdot h(y_1,y_2,y_3)$$

$$f_{Y_1,Y_2,Y_3}(y_1,y_2,y_3;\theta) = 6\cdot e^{-3\theta}\cdot e^{-(y_1+y_2+y_3)}$$

The term $e^{-(y_1+y_2+y_3)}$ depends on all three order statistics y_1, y_2, y_3 and not just on $Y_3 = \max(X_i)$

$\max(X_i)$ is not sufficient statistic for θ.

2019 Q#3 (42)

Ex In a sequence of "n" bernoulli trials with prob. of success "p", "x" successes were observed. Show that $\hat{p}(1-\hat{p})^2$ is not unbiased estimator of $p(1-p)^2$ but bias $\to 0$ as $n \to \infty$, where $\hat{p} = x/n$.

Sol.

$$\hat{p}(1-\hat{p})^2 = \hat{p}(1+\hat{p}^2-2\hat{p})$$
$$= \hat{p} + \hat{p}^3 - 2\hat{p}^2$$
$$E[\hat{p}(1-\hat{p})^2] = E(\hat{p}) + E(\hat{p}^3) - 2E(\hat{p}^2)$$
$$= E\left(\frac{x}{n}\right) + E\left(\frac{x}{n}\right)^3 - 2E\left(\frac{x}{n}\right)^2$$
$$= \frac{1}{n}E(x) + \frac{1}{n^3}E(x^3) - \frac{2}{n^2}E(x^2)$$

Now in a sequence of "n" bernoulli trials

$$E(x) = \boxed{np}, \quad E(x^2) = n(n-1)p^2 + np$$
$$E(x^3) = n(n-1)(n-2)p^3 + 3n(n-1)p^2 + np$$

$$E[\hat{p}(1-\hat{p})^2] = \frac{1}{n}(np) + \frac{1}{n^3}\left[(n^2-n)(n-2)p^3 + 3(n^2-n)p^2 + np\right]$$
$$- \frac{2}{n^2}\left[(n^2-n)p^2 + np\right]$$
$$= p + \frac{1}{n^3}\left[n^3p^3 - 2n^2p^3 - n^2p^3 + 2np^3 + 3n^2p^2 - 3np^2 + np\right]$$
$$- \frac{2}{n^2}\left[n^2p^2 - np^2 + np\right]$$
$$= p + \frac{1}{n^3}\left[n^3p^3 - 3n^2p^3 + 2np^3 + 3n^2p^2 - 3np^2 + np\right]$$
$$- \frac{2}{n^2}\left[n^2p^2 - np^2 + np\right]$$

(43)

$$= p + \left[p^3 \frac{(n^3 - 3n^2 + 2n)}{n^3} + 3p^2 \frac{(n^2 - n)}{n^3} + \frac{np}{n^3} \right]$$

$$- 2\left[p^2 \frac{(n^2 - n)}{n^2} + \frac{np}{n^2} \right]$$

$$= p + \left[p^3 \left(1 - \tfrac{3}{n} + \tfrac{2}{n^2}\right) + 3p^2 \left(\tfrac{1}{n} - \tfrac{1}{n^2}\right) + \tfrac{p}{n^2} \right]$$

$$- 2\left[p^2 (1 - \tfrac{1}{n}) + \tfrac{p}{n} \right] \quad \text{---} (i)$$

it is clear that $\hat{p}(1-\hat{p})^2$ is not an unbiased estimator of $p(1-p)^2$

For Bias

$$\text{Bias} = E(\hat{\theta}) - \theta$$

$$= \left\{ p + \left[p^3 \left(1 - \tfrac{3}{n} + \tfrac{2}{n^2}\right) + 3p^2\left(\tfrac{1}{n} - \tfrac{1}{n}\right) + \tfrac{p}{n^2} \right] - 2\left[p^2 (1 - \tfrac{1}{n}) + \tfrac{p}{n} \right] \right\} - p(1-p)^2$$

$$\lim_{n \to \infty} \text{Bias} = p + p^3 - 2p^2 - p(1-p)^2$$

$$= p(1 + p^2 - 2p) - p(1-p)^2$$

$$= p(1-p)^2 - p(1-p)^2$$

$$= pq^2 - pq^2 = 0$$

hence Bias $\to 0$ as $n \to \infty$

estimator of pq.

Example 1.6 Let $X_1, \ldots, X_n$ be a random sample of size n from the $N(0,\sigma^2)$. Show that t_1 and t_2 (defined below) are both unbiased estimators of σ.

$$t_1 = \left(\frac{1}{2}\sum_{i=1}^{n} X_i^2\right)^{1/2} \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n+1}{2}\right)},$$

2019 q#4

Solution $E(t_1) = \dfrac{\Gamma\left(\frac{n}{2}\right)}{\sqrt{2}\,\Gamma\left(\frac{n+1}{n}\right)} E\left(\sum_{i=1}^{n} X_i^2\right)^{1/2}$

$$= \frac{\Gamma\left(\frac{n}{2}\right)\sigma}{\sqrt{2}\,\Gamma\left(\frac{n+1}{n}\right)} E\left[\sum_{i=1}^{n} (X_i/\sigma)^2\right]^{1/2} \quad (1)$$

Since $\sum_{i=1}^{n} (X_i/\sigma)^2$ is distributed as chi-square with n d.f. (χ_n^2), we have

$$E\left[\sum_{i=1}^{n} (X_i/\sigma)^2\right]^{1/2} = E[\chi_n] = \int_0^\infty \chi f(\chi^2)\, d(\chi^2)$$

$$= \int_0^\infty \frac{\chi}{\Gamma\left(\frac{n}{2}\right) 2^{n/2}} e^{-\chi^2/2} (\chi^2)^{(n/2)-1}\, d(\chi^2)$$

Making the substitution $\chi^2/2 = y$, $d(\chi^2) = 2\,dy$, we get

$$= \frac{1}{\Gamma\left(\frac{n}{2}\right) 2^{n/2}} \int_0^\infty e^{-y} (2y)^{(n-1)/2}\, 2\,dy$$

$$= \frac{\sqrt{2}}{\Gamma\left(\frac{n}{2}\right)} \int_0^\infty e^{-y} y^{(n-1)/2}\, dy = \frac{\sqrt{2}}{\Gamma\left(\frac{n}{2}\right)} \Gamma\left(\frac{n+1}{2}\right) \quad (2)$$

$\frac{n}{2} + \frac{1}{2} - 1$

Making substitution from (2) into (1), we get

$E(t_1) = \sigma$, i.e. t_1 is an unbiased estimator of σ.

$n+1$

Q#5

DATE: ________ 2019 DAY: ________

State and prove Neyman Factorization:

(Statement) book pg 48

Let $X_1, X_2, \ldots, X_n$ be a random sample from a probability distribution with pdf $f(x;\theta)$ where θ is a parameter. A statistic $T(X_1, X_2 \ldots X_n)$ is sufficient for θ if and only if the joint pdf of the sample can be factored into the product of two functions:

$$f(x_1, x_2 \ldots, x_n; \theta) = g(T(x_1, x_2, \ldots x_n); \theta) \cdot h(x_1, x_2 \ldots x_n)$$

Proof:-

Necessity

Assume that $T(X_1, X_2 \ldots, X_n)$ is a sufficient statistic for θ. According to the definition of sufficiency, the conditional distribution of the sample given T should not depend on θ.

The joint pdf $f(x_1, x_2 \ldots x_n; \theta)$ using conditional and marginal probability of T:

$$f(x_1, x_2 \ldots, x_n; \theta) = f(x_1, x_2, \ldots x_n | T) \cdot f(T; \theta)$$

T is sufficient for θ.

The conditional p.d.f $f(x_1, x_2, \ldots, x_n | T)$ does not depend on θ.

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Therefore,

$$f(x_1, x_2, \dots x_n; \theta) = f(T;\theta) \cdot K(x_1, x_2 \dots x_n)$$

$$f(x_1, x_2, \dots x_n; \theta) = g(T(x_1, x_2 \dots x_n); \theta) \cdot h(x_1, x_2 \dots x_n)$$

Sufficiency:

Assume the joint p.d.f can be factored as:

$$f(x_1, x_2, \dots x_n; \theta) = g(T(x_1, x_2, \dots, x_n); \theta) \cdot h(x_1, x_2, \dots, x_n)$$

we need to show that $T(X_1, X_2, \dots, X_n)$ is sufficient for θ. This means that conditional pdf $f(x_1, x_2, \dots x_n / T)$ should not depend on θ.

using bayes theorem:

$$f(x_1, x_2, \dots, x_n / T = t; \theta) = \frac{f(x_1, x_2, \dots x_n; \theta)}{f(T = t; \theta)}$$

$$f(x_1, x_2, \dots, x_n; \theta) = g(T(x_1, x_2, \dots, x_n); \theta) \cdot h(x_1, x_2, \dots, x_n)$$

$$f(T = t; \theta) = \int_{T=t} g(T(x_1, x_2 \dots x_n); \theta) \cdot h(x_1, x_2 \dots x_n) \, dx_1 dx_2 \dots dx_n$$

$$f(x_1, x_2 \dots x_n / T = t; \theta) = \frac{g(T(x_1, x_2 \dots x_n); \theta) \cdot h(x_1, x_2 \dots x_n)}{\int_{T=t} g(T(x_1, x_2 \dots, x_n); \theta) \cdot h(x_1, x_2 \dots, x_n) \, dx_1 dx_2 \dots dx_n}$$

$h(x_1, x_2 \dots x_n)$ does not depend on θ.

proving that T is sufficient for θ.

2018, 2019

DATE: Q#4, Q#6 DAY:

Given the p.d.f:

$f(X;\theta) = e^{-(x-\theta)}$ for $\theta < x < \infty$

for a random X_1, X_2, X_3 the joint p.d.f is:

$L(\theta; X_1, X_2, X_3) = f(X_1;\theta)\cdot f(X_2;\theta)\cdot f(X_3;\theta)$

$L(\theta; X_1, X_2, X_3) = e^{-(X_1-\theta)} \cdot e^{-(X_2-\theta)} \cdot e^{-(X_3-\theta)}$

$L\theta; X_1, X_2, X_3) = e^{-(X_1+X_2+X_3-3\theta)}$

order the Sample values as $Y_1 < Y_2 < Y_3$

where $Y_3 = \max(X_i)$.

The joint p.d.f order statistics given as:

$f_{Y_1,Y_2,Y_3}(y_1, y_2, y_3;\theta) = 3!\cdot e^{-(y_1-\theta)} \cdot e^{-(y_2-\theta)} \cdot e^{-(y_3-\theta)}$

$f_{Y_1,Y_2,Y_3}(y_1, y_2, y_3;\theta) = 6\cdot e^{-(y_1+y_2+y_3-3\theta)}$

$f_{Y_1,Y_2,Y_3}(y_1, y_2, y_3;\theta) = 6\cdot e^{-3\theta} \cdot e^{-(y_1+y_2+y_3)}$

By using factorization theorem:

$T(X)$ is sufficient for θ.

The joint p.d.f can be factored into:

$L(\theta; X) = g(T(X),\theta)\cdot h(X)$

$T(X) = \max(X_i) = Y_3$

For $Y_3 = \max(X_i)$:

$f_{Y_3}(y_3;\theta) = \int_{-\infty}^{y_3}\int_{-\infty}^{y_3}\int_{-\infty}^{y_3} 6\cdot e^{-3\theta}\cdot e^{-(y_1+y_2+y_3)}\, dy_1\, dy_2\, dy_3$

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$f_{Y_1, Y_2, Y_3}(y_1, y_2, y_3; \theta) = g(Y_3, \theta) \cdot h(y_1, y_2, y_3)$

$f_{Y_1, Y_2, Y_3}(y_1, y_2, y_3; \theta) = 6 \cdot e^{-3\theta} \cdot e^{-(y_1 + y_2 + y_3)}$

The term $e^{-(y_1+y_2+y_3)}$ depends on all three order statistics y_1, y_2, y_3 and not just on $Y_3 = \max(X_i)$

$\max(X_i)$ is not sufficient statistic for θ.

2020

Q4:- Prove that minimum variance bound estimator is Unique. Repeat in 2018 Q#6 (10)

Q5:- State ~~any~~ and prove Rao-Blackwell theorem. (10)

p.d.f Repeat in 2017

Let $X_1, X_2, \ldots, X_n$ be a random sample from uniform distribution $U(\theta, \theta+1)$ and $Y_1 < Y_2 < Y_3$ be the order statistic of order 3, then Y_1 check for possible unbiasedness.

2020 q#3

$$f(x;\theta) = \frac{1}{\cancel{\theta}+1-\cancel{\theta}} = 1$$

$$F(x) = \int_{-\infty}^{x} f(x)\,dx = \int_{\theta}^{x} dx = |x|_{\theta}^{x} = x - \theta$$

$$\boxed{F(y_1) = y_1 - \theta}$$

$$g(y_1) = n[1-F(y_1)]^{n-1} f(y_1)$$

$$g(y_1) = n[1-y_1+\theta]^{n-1} \cdot 1$$

$$E[Y_1] = \int_{\theta}^{\theta+1} y_1 \cdot g(y_1)\,dy_1$$

$$= n\int_{\theta}^{\theta+1} y_1(1-y_1+\theta)^{n-1}\,dy_1$$

$$= 3\int_{\theta}^{\theta+1} y_1(1-y_1+\theta)^2\,dy_1$$

$$\cancel{= 3\int_0^1 (\theta-z)\ldots}$$

$$= 3\int_0^1 (\theta+z)(1-z)^2\,dz$$

$$= 3\left[\theta\int_0^1 (1-z)^2\,dz + \int_0^1 z(1-z)^2\,dz\right]$$

$$= 3\left[\theta\int_0^1 u^2\,du + B(2,3)\right]$$

$$= 3\left[\theta\left|\frac{u^3}{3}\right|_0^1 + B(2,3)\right]$$

$$= \theta + 3\frac{\Gamma(2)\Gamma(3)}{\Gamma(2+3)} = \theta + 3\frac{1!\,2!}{4!} = \theta + \frac{\cancel{2}\,\cancel{3}}{4\times\cancel{3}\times\cancel{2}} = \theta + \frac{1}{4}$$

~~$-y_1+\theta = z_1$~~

~~$y_1 = (\theta - z_1)$~~

~~$y_1 = \theta;\ z_1 = 0$~~

~~$y_1 = \theta+1;\ z_1 = \theta-1+\theta$~~

~~$z_1 = -1$~~

$y_1 - \theta = z$

$y_1 = \theta + z$

$y_1 = \theta;\ z = 0$

$y_1 = \theta+1;\ z = \cancel{\theta}+1-\cancel{\theta} = 1$

$dy_1 = dz$

$1-z = u \quad -dz = du$

$z = 0;\ u = 1.$

$z = 1;\ u = 0$

2021 q#1

Ex: Show that in samples from the Gamma dist^n having pdf: $f(x;\theta,p) = \frac{1}{\Gamma p\, \theta^p} x^{p-1} e^{-x/\theta}$, $0 \le x \le \infty, p>0$

the MVB estimator of θ for known p is $\bar{x}/p$ with variance $\frac{\theta^2}{np}$, while if $\theta = 1$, that of $\frac{\partial}{\partial p} \ln \Gamma p$ is $\frac{1}{n}\sum_{i=1}^{n} \ln x_i$ with variance $\frac{\frac{\partial^2}{\partial p^2} \ln \Gamma p}{n}$

Sol: the LHF is given as

$$L = (\Gamma p)^{-n}\, \theta^{-np} \,(\Pi x_i)^{p-1} e^{-\Sigma x_i/\theta}$$

taking log on both sides

$$\log L = -n \log \Gamma p - np \log \theta + (p-1)\log(\Pi x_i) - \Sigma x_i \theta^{-1} \log e \quad \text{---(i)}$$

$$\frac{\partial \log L}{\partial \theta} = 0 - \frac{np}{\theta} + 0 - \Sigma x_i \theta^{-2}(-1)$$

$$= \frac{\Sigma x_i}{\theta^2} - \frac{np}{\theta} = \frac{\Sigma x_i}{\theta^2}\frac{np}{np} - \frac{np}{\theta}\cdot\frac{\theta}{\theta}$$

$$= \frac{np}{\theta^2}\left[\frac{\Sigma x_i}{np} - \theta\right] = \frac{np}{\theta^2}\left[\bar{x}/p - \theta\right]$$

comparing with $\frac{\partial \log L}{\partial \theta} = A(\theta)[\theta^* - h(\theta)]$

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$A(\theta) = np/\theta^2$, $\theta^* = \bar{x}/p$, $h(\theta) = \theta$

hence $\bar{x}/p$ is MVB estimator of θ with variance

$$V(\theta^*) = \frac{1}{A(\theta)} = \frac{1}{np/\theta^2} = \theta^2/np$$

Now put $\theta = 1$ in eq (i)

$$\log L = -n \log \Gamma p - np \log(1) + (p-1) \log(\Pi x_i) - \Sigma x_i$$

$$= -n \log \Gamma p + (p-1) \Sigma \log x_i - \Sigma x_i$$

$$= -n \log \Gamma p + p \Sigma \log x_i - \Sigma \log x_i - \Sigma x_i$$

$$\frac{\partial \log L}{\partial p} = -n \frac{\partial}{\partial p} \log \Gamma p + \Sigma \log x_i - 0 - 0$$

$$= \Sigma \log x_i - n \frac{\partial}{\partial p} \log \Gamma p$$

$$= n\left[\frac{\Sigma \log x_i}{n} - \frac{\partial}{\partial p} \log \Gamma p\right]$$

$A(\theta) = n$, $\theta^* = \frac{\Sigma \log x_i}{n}$, $h(\theta) = \frac{\partial}{\partial p} \log \Gamma p$

with variance $V(\theta^*) = \frac{h'(\theta)}{A(\theta)}$

$$h(\theta) = \frac{\partial}{\partial p} \log \Gamma p \Rightarrow h'(\theta) = \frac{\partial^2}{\partial p^2} \log \Gamma p$$

$$V(\theta^*) = \frac{\frac{\partial^2 \log \Gamma p}{\partial p^2}}{n} = \frac{\partial^2}{\partial p^2} \log \Gamma p \Big/ n$$

$$= \frac{\partial^2 \log \Gamma p}{n\, \partial p^2}$$

$i=1$

2021 q#2

Theorem 2.3 The sufficient statistic is unique.

Proof If there are two distinct sufficient statistics t_1 and t_2, then we have

$$f(t_1,t_2;\theta) = h_1(t_1;\theta)\, k_1(t_2 \mid t_1) = h_2(t_2;\theta)\, k_2(t_1 \mid t_2)$$

or $$h_1(t_1;\theta) = h_2(t_2;\theta) \cdot k_2(t_1 \mid t_2)\, k_1(t_2 \mid t_1)$$

Writing $h_3(t_1,t_2) = k_2(t_1 \mid t_2)\, k_1(t_2 \mid t_1)$, we obtain

$$h_1(t_1;\theta) = h_2(t_2;\theta)\, h_3(t_1,t_2)$$

identically in θ. This cannot hold unless t_1 and t_2 are functionally related.

For some problems, no single sufficient statistic exists. However, there will always exist *jointly sufficient statistics* as defined below.

Definition 2.2 Let $X_1,\ldots,X_n$ be a random sample from the p.d.f. $f(x;\theta)$. The statistics $T_1,\ldots,T_r$ are defined to be jointly sufficient if and only if the conditional distribution of $X_1,\ldots,X_n$ given $T_1 = t_1,\ldots,T_r = t_r$ does not depend on θ. (The parameter θ can be a vector).

The sample $X_1,\ldots,X_n$ itself is always jointly sufficient since the conditional distribution of the sample given the sample does not depend on θ. Also, the order statistics $Y_1,\ldots,Y_n$ are jointly sufficient for random sampling. If the order statistics are $(Y_1 = y_1,\ldots,Y_n = y_n)$, then the only values that can be taken on by $(X_1,\ldots,X_n)$ are the permutations of $y_1,\ldots,y_n$. Since the sampling is random, each of the $n!$ permutations is equally likely. Thus, given the values of the order statistics, the probability that the sample equals a particular permutation of these given values of the order statistics is $1/n!$, which is independent of θ.

Theorem 2.4 If $T_1 = t_1(X_1,\ldots,X_n),\ldots,T_r = t_r(X_1,\ldots,X_n)$ is a set of jointly sufficient statistics, then any set of one-to-one functions, or transformations, of $T_1,\ldots,T_r$ is also jointly sufficient.

For example, if ΣX_i and ΣX_i^2 are jointly sufficient, then $\bar{X}$ and $\Sigma(X_i - \bar{X})^2 = \Sigma X_i^2 - n\bar{X}^2$ are also jointly sufficient. However, $\bar{X}^2$ and $\Sigma(X_i - \bar{X})^2$ are not jointly sufficient since they are not one-to-one functions of ΣX_i and ΣX_i^2.

Theorem 2.5 Factorization Theorem (Jointly Sufficient Statistics) Let $X_1,\ldots,X_n$ be a random sample of size n from the p.d.f. $f(x;\theta)$, where the parameter θ may be a vector. A set of statistics $T_1 = t_1(X_1,\ldots,X_n),\ldots,T_r = t_r(X_1,\ldots,X_n)$ is jointly sufficient if and only if the joint distribution of $X_1,\ldots,X_n$ can be written as

$$\begin{aligned} f(x_1,\ldots,x_n;\theta) &= h[t_1(x_1,\ldots,x_n),\ldots,t_r(x_1,\ldots,x_n);\theta]\, k(x_1,\ldots,x_n) \\ &= h(t_1,\ldots,t_r;\theta)\, k(x_1,\ldots,x_n) \end{aligned}$$

where the function $k(x_1,\ldots,x_n)$ is nonnegative and does not involve the parameter θ and the function $h(t_1,\ldots,t_r;\theta)$ is nonnegative and depends on $x_1,\ldots,x_n$ only through the functions $t_1(x_1,\ldots,x_n),\ldots,t_r(x_1,\ldots,x_n)$.

Theorem 30. Imp

Let $x_1, x_2, \ldots\ldots, x_n$ is a random sample from a distribution $U(0,a)$. Then $\hat{M} = \max(x_1, \ldots\ldots x_n)$ check it for unbiasedness and consistency.

Proof.

$$f(x) = \frac{1}{a-0} = \frac{1}{a}$$

$\boxed{f(x) = \frac{1}{b-a}}$ function of uniform dist

$$F(x) = \int_0^x f(x)\,dx$$

$$= \int_0^x \frac{1}{a}\,dx$$

$$= \frac{1}{a}\int_0^x 1\,dx$$

$$= \frac{1}{a}\,\big|x\big|_0^x$$

$$= \frac{1}{a}(x-0)$$

$$F(x) = \frac{x}{a}$$

The distribution of max order statistic can be written as:

$$g(y_n) = n\left[F(y_n)\right]^{n-1} f(y_n) \text{ ——— (i)}$$

$$f(x) = f(y_n) = f(y) = \frac{1}{a}$$

$$F(x) = F(y_n) = f(y) = \frac{y_n}{a} = \frac{y}{a}$$

put in eq (i)

$$g(y_n) = g(y) = n\left(\frac{y}{a}\right)^{n-1}\frac{1}{a}$$

$$= \frac{n}{a\,a^{n-1}}\,y^{n-1}$$

$$g(y) = \frac{n}{a^n}\,y^{n-1} \qquad 0<y<a$$

For unbiasedness

$$E(\hat{M}) = E(y_n) = E(y)$$

$$= \int_{-\infty}^{+\infty} y\,g(y)\,dy$$

$$= \int_0^a y\cdot\frac{n}{a^n}\,y^{n-1}\,dy$$

$$= \frac{n}{a^n}\int_0^a y\,y^{n-1}\,dy$$

$$= \frac{n}{a^n}\int_0^a y^n\,dy$$

$$= \frac{n}{a^n}\left|\frac{y^{n+1}}{n+1}\right|_0^a$$

$$= \frac{n}{a^n(n+1)}\left[a^{n+1}-0\right]$$

$$= \frac{n}{a^n(n+1)}\left(a^{n+1}\right)$$

$$E(\hat{m}) = \frac{n\,\cancel{a^n}\cdot a}{\cancel{a^n}(n+1)} = \frac{na}{n+1}$$

$$E(\hat{m}) = E(y_n)$$

$$= \frac{na}{n+1} \neq a$$

It is biased

$$\text{Bias} = E(\hat{\varepsilon}) - \sigma$$

$$= \frac{na}{n+1} - a$$

$$= a\left[\frac{n}{n+1} - 1\right]$$

$$\text{Bias} = a\left[\frac{n/n}{(n+1)/n} - 1\right]$$

$$= a\left[\frac{1}{1+\frac{1}{n}} - 1\right]$$

Apply Limit

$$\lim_{n\to\infty} \text{Bias} = a\left[\frac{1}{1+\frac{1}{\infty}} - 1\right]$$

$$= a(1-1)$$

$$= 0$$

$\hat{M}$ is asymptotically unbiased.

Now, For Consistency

$$V(\hat{M}) = V(y_n)$$

$$= E\left[y_n^2\right] - \left[E(y_n)\right]^2 \text{ ——— (ii)}$$

$$E(y_n^2) = \int y_n^2\,g(y_n)\,dy_n$$

$$= \int_0^a y^2\,\frac{n}{a^n}\,y^{n-1}\,dy$$

$$= \frac{n}{a^n}\int_0^a y^{n+1}\,dy$$

$$= \frac{n}{a^n}\left|\frac{y^{n+2}}{n+2}\right|_0^a$$

$$= \frac{n\,a^{n+2}}{a^n(n+2)} = \frac{n\,\cancel{a^n}\cdot a^2}{\cancel{a^n}(n+2)}$$

$$E(y_n^2) = \frac{na^2}{n+2}$$

put in eq (i)

$$V(\hat{m}) = \left[\frac{na^2}{n+2} - \left(\frac{na}{n+1}\right)^2\right]$$

$$= \frac{na^2}{n+2} + \frac{n^2a^2}{(n+1)^2}$$

$$V(\hat{m}) = na^2\left[\frac{1}{n+2} - \frac{n}{n^2+1+2n}\right]$$

$$= na^2\left[\frac{\cancel{n^2}+1+\cancel{2n}-\cancel{n^2}-\cancel{2n}}{(n+2)(n^2+1+2n)}\right]$$

$$= \frac{na^2}{(n+2)(n^2+1+2n)} = \frac{na^2}{n^3+n+2n^2+2n^2+2+4n}$$

$$= \frac{na^2}{n^3+4n^2+5n+2}$$

$$V(\hat{m}) = \frac{a^2}{n^2+4n+5+\frac{2}{n}}$$

Apply Limit

$$\lim_{n\to\infty} V(\hat{m}) = \frac{a^2}{\infty}$$

$$= 0$$

$\hat{m}$ is a Consistent Proved

Theorem 31

Let $x_1, \ldots, x_n$ be a r.s from $U(0, a)$

a) Check whether $\hat{m} = \min(x_1, \ldots, x_n)$ unbiased or Biased.

b) If Biased find amount of Bias.

c) It is asymptotically unbiased

d) It is consistent estimator.

Proof

$$f(x) = \frac{1}{a-0} = \frac{1}{a}$$

$$F(x) = \int_0^x f(x)\,dx$$

$$= \int_0^x \frac{1}{a}\,dx$$

$$= \frac{1}{a}\int_0^x 1\,dx$$

$$= \frac{1}{a}\left|x\right|_0^x$$

$$= \frac{1}{a}(x-0)$$

$$F(x) = \frac{x}{a}$$

P.d.f of minimum value is :

$$g(y_1) = n f(y_1)\left[1 - F(y_1)\right]^{n-1} \quad \text{--- (1)}$$

$$f(x) = f(y_1) = \frac{1}{a}$$

$$F(x) = F(y_1) = f(y) = \frac{y_1}{a} = \frac{y}{a}$$

Put in eq (1)

$$g(y_1) = g(y) = n\left(\frac{1}{a}\right)\left[1 - \frac{y}{a}\right]^{n-1}$$

$$= \frac{n}{a}\left(1 - \frac{y}{a}\right)^{n-1}$$

a For unbiasedness

$$E(\hat{m}) = E(y_1) = E(y)$$

$$= \int_{-\infty}^{\infty} y\, g(y)\,dy$$

$$= \int_0^a y \cdot \frac{n}{a}\left(1 - \frac{y}{a}\right)^{n-1} dy$$

$$= \frac{n}{a}\int_0^a y\left(1 - \frac{y}{a}\right)^{n-1} dy$$

$$= \frac{n}{a}\int_0^a y\left(\frac{a-y}{a}\right)^{n-1} dy$$

$$= \frac{n}{a\,a^{n-1}}\int_0^a y(a-y)^{n-1}\,dy$$

$$E(\hat{m}) = \frac{n}{a^n}\int_0^a y(a-y)^{n-1}\,dy$$

put $a - y = z$

$a - z = y$

$-dz = dy$

when $y = 0$ then $z = a$

when $y = a$ then $z = 0$

$$E(\hat{m}) = \frac{n}{a^n}\int_a^0 (a-z)\,z^{n-1}(-dz)$$

$$= \frac{n}{a^n}\int_0^a (a-z)\,z^{n-1}\,dz$$

$$= \frac{n}{a^n}\int_0^a \left(a z^{n-1} - z\,z^{n-1}\right)dz$$

$$= \frac{n}{a^n}\int_0^a \left(a z^{n-1} - z^n\right)dz$$

$$= \frac{n}{a^n}\left[\int_0^a a z^{n-1}\,dz - \int_0^a z^n\,dz\right]$$

$$= \frac{n}{a^n}\left[a\int_0^a z^{n-1}\,dz - \int_0^a z^n\,dz\right]$$

$$E(\hat{m}) = \frac{n}{a^n}\left[a\left|\frac{z^{n-1+1}}{n-1+1}\right|_0^a - \left|\frac{z^{n+1}}{n+1}\right|_0^a\right]$$

$$= \frac{n}{a^n}\left[\frac{a}{n}(a^n - 0) - \frac{1}{n+1}(a^{n+1} - 0)\right]$$

$$= \frac{n}{a^n}\left[\frac{a\,a^n}{n} - \frac{a^{n+1}}{n+1}\right]$$

$$= \frac{n}{a^n}\left[\frac{a^{n+1}}{n} - \frac{a^{n+1}}{n+1}\right]$$

$$= \frac{n\,a^{n+1}}{a^n}\left[\frac{1}{n} - \frac{1}{n+1}\right]$$

$$E(\hat{m}) = \frac{n\,a^{n+1}}{a^n}\left[\frac{\cancel{n}+1-\cancel{n}}{n(n+1)}\right]$$

$$= \frac{\cancel{n}\,a^{n+1}}{a^n}\left[\frac{1}{\cancel{n}(n+1)}\right]$$

$$= \frac{a^n a}{a^n}\left(\frac{1}{n+1}\right)$$

$$E(\hat{m}) = \frac{a}{n+1} \neq a$$

This shows that $\hat{m}$ is biased estimator.

b Amount of bias

$$\text{Bias} = E(\hat{\theta}) - \theta$$

$$= E(\hat{m}) - a$$

$$\text{Bias} = \frac{a}{n+1} - a$$

$$= \frac{\cancel{a} - an - \cancel{a}}{n+1}$$

$$= \frac{-an}{n+1}$$

$$\text{Bias} = \frac{-a}{1+\frac{1}{n}} \quad (\text{-ve Bias})$$

c Apply Limit

$$\lim_{n\to\infty} \text{Bias} = \frac{-a}{1+\frac{1}{\infty}}$$

$$= \frac{-a}{1+0}$$

$$= -a \neq 0$$

This show that $\hat{m}$ is not asymptotically unbiased estimator.

d

$$V(\hat{m}) = E(y_1^2) - \left[E(y_1)\right]^2 \quad \text{--- (A)}$$

$$E(y_1^2) = \int y_1^2\, g(y_1)\,dy_1$$

$$= \int_0^a y^2\,\frac{n}{a}\left(1 - \frac{y}{a}\right)^{n-1} dy$$

$$= \frac{n}{a}\int_0^a y^2\left(1 - \frac{y}{a}\right)^{n-1} dy$$

$$= \frac{n}{a}\int_0^a y^2\left(\frac{a-y}{a}\right)^{n-1} dy$$

$$= \frac{n}{a\,a^{n-1}}\int_0^a y^2(a-y)^{n-1}\,dy$$

$$E(y_1^2) = \frac{n}{a^n}\int_0^a y^2(a-y)^{n-1}\,dy$$

put $a - y = z$

$a - z = y$

$-dz = dy$

when $y = 0$ then $z = a$

when $y = a$ then $z = 0$

$$E(y_1^2) = \frac{n}{a^n}\int_a^0 (a-z)^2 z^{n-1}(-dz)$$

$$= \frac{n}{a^n}\int_0^a (a-z)^2 z^{n-1}\,dz$$

$$= \frac{n}{a^n}\int_0^a (a^2 + z^2 - 2az)\,z^{n-1}\,dz$$

$$= \frac{n}{a^n}\int_0^a \left(a^2 z^{n-1} + z^{n+1} - 2a z^n\right)dz$$

$$= \frac{n}{a^n}\left[\int_0^a a^2 z^{n-1}\,dz + \int_0^a z^{n+1}\,dz - 2\int_0^a a z^n\,dz\right]$$

$$= \frac{n}{a^n}\left[a^2\int_0^a z^{n-1}\,dz + \int_0^a z^{n+1}\,dz - 2a\int_0^a z^n\,dz\right]$$

$$= \frac{n}{a^n}\left[a^2\left|\frac{z^n}{n}\right|_0^a + \left|\frac{z^{n+2}}{n+2}\right|_0^a - 2a\left|\frac{z^{n+1}}{n+1}\right|_0^a\right]$$

$$= \frac{n}{a^n}\left[\frac{a^2}{n}(a^n - 0) + \frac{1}{n+2}(a^{n+2} - 0) - \frac{2a}{n+1}(a^{n+1} - 0)\right]$$

$$= \frac{n}{a^n}\left[\frac{a^2 a^n}{n} + \frac{a^{n+2}}{n+2} - \frac{2a\,a^{n+1}}{n+1}\right]$$

$$= \frac{n}{a^n}\left[\frac{a^{n+2}}{n} + \frac{a^{n+2}}{n+2} - \frac{2a^{n+2}}{n+1}\right]$$

$$= \frac{n\,a^{n+2}}{a^n}\left[\frac{1}{n} + \frac{1}{n+2} - \frac{2}{n+1}\right]$$

$$= \frac{n\,a^{n+2}}{a^n}\left[\frac{(n+1)(n+2) + n(n+1) - 2n(n+2)}{n(n+1)(n+2)}\right]$$

$$= \frac{\cancel{n}\,a^{n+2}}{a^n}\left[\frac{\cancel{n^2} + \cancel{2n} + \cancel{n} + 2 + \cancel{n^2} + \cancel{n} - \cancel{2n^2} - \cancel{4n}}{\cancel{n}(n+1)(n+2)}\right]$$

$$E(y_1^2) = \frac{a^{n+2}}{a^n}\left[\frac{2}{(n+1)(n+2)}\right]$$

$$E(y_1^2) = \frac{2a^{n+2}}{a^n(n+1)(n+2)}$$

$$E(y_1^2) = \frac{2a^n a^2}{a^n(n+1)(n+2)}$$

$$= \frac{2a^2}{(n+1)(n+2)}$$

put in eq (A)

$$V(\hat{m}) = \frac{2a^2}{(n+1)(n+2)} - \left(\frac{a}{n+1}\right)^2$$

$$= \frac{2a^2}{(n+1)(n+2)} - \frac{a^2}{(n+1)^2}$$

$$= \frac{a^2}{n+1}\left[\frac{2}{n+2} - \frac{1}{n+1}\right]$$

$$= \frac{a^2}{n+1}\left[\frac{2(n+1) - (n+2)}{(n+1)(n+2)}\right]$$

$$= \frac{a^2}{n+1}\left[\frac{2n + \cancel{2} - n - \cancel{2}}{(n+1)(n+2)}\right]$$

$$= \frac{a^2}{n+1}\left[\frac{n}{(n+1)(n+2)}\right]$$

$$V(\hat{m}) = \frac{n\,a^2}{(n+1)^2(n+2)}$$

$$= \frac{n\,a^2}{(n^2 + 1 + 2n)(n+2)}$$

$$= \frac{n\,a^2}{n^3 + n + 2n^2 + 2n^2 + 2 + 4n}$$

$$= \frac{n\,a^2}{n^3 + 4n^2 + 5n + 2}$$

$$V(\hat{m}) = \frac{a^2}{\ldots + 2/n}$$

Apply

$$\lim_{n\to\infty} V(\hat{m}) = \frac{a^2}{\infty} = 0$$

This shows that $\hat{m}$ is consistent estimator.

$E(2x) = 2E(x) = \dots$

2022 q#3

Q. Let $Y_1 < Y_2 < Y_3$ be the order statistics of a random sample of size 3 from pdf $f(x;\theta) = \frac{1}{\theta}$, $0<x<\theta$, $0<\theta<\infty$. Show that $4Y_1$, $2Y_2$ and $\frac{4}{3}Y_3$ are all unbiased estimator of θ with respective variances $\frac{3\theta^2}{5}$, $\frac{\theta^2}{5}$ and $\frac{\theta^2}{15}$.

The pdf of r^{th} order statistic Y_r from a random sample of size 'n' is

$$g(y_r) = \frac{n!}{(r-1)!\,(n-r)!}\left[F(y_r)\right]^{r-1}\left[1-F(y_r)\right]^{n-r} f(y_r) \quad \text{—— (1)}$$

where $r = 1,2,3$, and $n=3$

$$F(x \le a) = \int_0^a f(x)\,dx$$

$$\int_0^x \frac{1}{\theta}\,dx = \frac{1}{\theta}|x|_0^x = \frac{x}{\theta}$$

$$F(y_r) = \frac{y_r}{\theta} \qquad F(y_1) = \frac{y_1}{\theta}$$

Put $r=1$ in (1), $n=3$

$$g(y_1) = \frac{3!}{0!\times 2!}\left[\frac{y_1}{\theta}\right]^{1-1}\left[1-\frac{y_1}{\theta}\right]^{3-1}\frac{1}{\theta}$$

$$g(y_1) = \frac{3\times\cancel{2}}{1\times\cancel{2}\times 1}\left[1-\frac{y_1}{\theta}\right]^2\frac{1}{\theta} = \frac{3}{\theta}\left[1-\frac{y_1}{\theta}\right]^2$$

$$E(Y_1) = \int_0^\theta y_1\, f(y_1)\,dy_1$$

$$= \int_0^\theta y_1\,\frac{3}{\theta}\left[1-\frac{y_1}{\theta}\right]^2 dy_1$$

$$= 3\int_0^\theta \frac{y_1}{\theta}\left[1-\frac{y_1}{\theta}\right]^2 dy_1$$

$$= 3\int_0^1 z(1-z)^2\,\theta\,dz$$

$$= 3\theta\int_0^1 z\,(1-z)^2\,dz$$

put $\frac{y_1}{\theta} = z$

$y_1 = \theta z$

$dy_1 = \theta\,dz$

$y_1 \to 0;\ z\to 0$

$y_1 \to \theta;\ z\to 1$

$\because \beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$ {beta function}.

$$E(Y_1) = 3\theta\,\beta(2,3) = 3\theta\,\frac{\Gamma(2)\Gamma(3)}{\Gamma 5} = 3\theta\,\frac{(2-1)!\,(3-1)!}{(5-1)!}$$

$$= 3\theta\,\frac{1!\,2!}{4!} = \frac{\cancel{3}\theta\ 1\cdot\cancel{2!}}{4\times\cancel{3}\times\cancel{2!}} = \frac{\theta}{4}$$

$E[4Y_1] = 4E[Y_1] = 4\,\frac{\theta}{4} = \theta$ unbiased

For variance: $Var[4Y_1] = 4^2\,Var[Y_1] = 16\,Var[Y_1] \longrightarrow (b)$

$$Var(Y_1) = E[Y_1^2] - (E[Y_1])^2$$

$$= E[Y_1^2] - \left(\frac{\theta}{4}\right)^2 \longrightarrow (c)$$

$$E[Y_1^2] = \int_0^\theta y_1^2\, g(y_1)\,dy_1 = \int_0^\theta y_1^2\, 3\left(1-\frac{y_1}{\theta}\right)^2\frac{1}{\theta}\,dy_1$$

$$= \frac{3}{\theta}\int_0^\theta y_1^2\left(1-\frac{y_1}{\theta}\right)^2 dy_1$$

$$= \frac{3}{\theta}\int_0^1 \theta^2 z^2(1-z)^2\,\theta\,dz$$

$$= 3\theta^2\int_0^1 z^2(1-z)^2\,dz$$

$$= 3\theta^2\int_0^1 z^{3-1}(1-z)^{3-1}\,dz$$

$$= 3\theta^2\,\beta(3,3) = 3\theta^2\,\frac{\Gamma(3)\Gamma(3)}{\Gamma 6} = 3\theta^2\,\frac{2!\,2!}{5!}$$

$$= \cancel{3}\theta^2\,\frac{\cancel{2!}\ 2\times 1}{5\times 4\times\cancel{3}\times\cancel{2!}} = \frac{\theta^2}{10}$$

put $\frac{y_1}{\theta} = z$

$y_1 = \theta z$

$dy_1 = \theta\,dz$

$y_1\to 0;\ z\to 0$

$y_1\to\theta;\ z\to 1$

$2|10-16$
$\ |5-8$

$$Var(Y_1) = \frac{\theta^2}{10} - \frac{\theta^2}{16} = \frac{8\theta^2 - 5\theta^2}{80} = \frac{3\theta^2}{80}$$

Put in eq (b)

$$Var[4Y_1] = 16\,\frac{3\theta^2}{80} = \frac{3}{5}\,\theta^2.$$

q#5

Let t=x1+x2+x3 and k=x1+2X2+x3 be two statistics where xi follow the bernoulli distribution, check whether or not the 't' and 'k' are sufficient by using conditional probability approach.

To check whether the statistics $t = x_1 + x_2 + x_3$ and $k = x_1 + 2x_2 + x_3$ are sufficient for the parameter of interest, we'll use the concept of sufficiency via the Neyman-Fisher factorization theorem and conditional probability.

1. Setup

Given that x_i follows a Bernoulli distribution: $X_i \sim \text{Bernoulli}(p)$ The probability mass function for each x_i is:
$P(X_i = x_i) = p^{x_i}(1-p)^{1-x_i}, \quad x_i \in \{0, 1\}$

2. Joint Distribution

The joint probability mass function of X_1, X_2, X_3 is:
$P(X_1 = x_1, X_2 = x_2, X_3 = x_3) = p^{x_1+x_2+x_3}(1-p)^{3-(x_1+x_2+x_3)}$

3. Sufficiency of

To check if $t = x_1 + x_2 + x_3$ is sufficient for p, we use the Neyman-Fisher factorization theorem. According to the theorem, a statistic $T(X)$ is sufficient for parameter θ if and only if the joint probability mass function can be factorized into two non-negative functions:
$P(x_1, x_2, x_3 \mid \theta) = g(T(x); \theta)h(x)$

For $t = x_1 + x_2 + x_3$:
$P(X_1 = x_1, X_2 = x_2, X_3 = x_3 \mid t) = \frac{P(X_1 = x_1, X_2 = x_2, X_3 = x_3)}{P(T = t)}$

Given that $t = x_1 + x_2 + x_3$, the joint probability mass function can be written as:
$P(X_1 = x_1, X_2 = x_2, X_3 = x_3) = p^t(1-p)^{3-t}$

This factorizes into:
$P(X_1, X_2, X_3 \mid t) = p^t(1-p)^{3-t} \times (\text{constant})$

Since the factorization depends only on t and the constants, $t = x_1 + x_2 + x_3$ is sufficient for p.

4. Sufficiency of

For $k = x_1 + 2x_2 + x_3$:

Let's check if k can be factorized similarly. The joint probability mass function in terms of k is not straightforward, so we use the conditional probability approach.

Consider the conditional distribution of $(X_1, X_2, X_3) \mid k$: $P(x_1, x_2, x_3 \mid k) = \frac{P(x_1, x_2, x_3)}{P(K = k)}$

Since $k = x_1 + 2x_2 + x_3$, the joint probability mass function can be written as:
$P(x_1, x_2, x_3) = p^{x_1+x_2+x_3}(1-p)^{3-(x_1+x_2+x_3)}$

It is not easy to see how this factorizes directly in terms of k. Therefore, we check whether knowing k is sufficient to uniquely determine the value of t.

1. When $k = 0$, $x_1 = 0, x_2 = 0, x_3 = 0$ implies $t = 0$.
2. When $k = 1$, $x_1 = 1, x_2 = 0, x_3 = 0$ implies $t = 1$.
3. When $k = 2$, $(x_1, x_2, x_3) = (0, 1, 0)$ or $(1, 0, 1)$ implies t can be 1 or 2.
4. When $k = 3$, similar ambiguity exists.

Since k does not uniquely determine t, it cannot be sufficient for p.

Conclusion:

- The statistic $t = x_1 + x_2 + x_3$ is sufficient for p.
- The statistic $k = x_1 + 2x_2 + x_3$ is not sufficient for p.

If you have more questions or need further efficient estimator.

2022 q#2

Statement:

Let θ be a parameter of interest, and let $\hat{\theta}$ be an estimator of θ. If $\hat{\theta}$ is an efficient estimator, then it is unique in the class of unbiased estimators that achieve the Cramér-Rao lower bound.

Proof:

Consider the following steps to prove the Uniqueness Theorem for Efficiency.

1. **Cramér-Rao Lower Bound (CRLB)**: The Cramér-Rao lower bound provides a lower bound on the variance of any unbiased estimator of a parameter θ. If $I(\theta)$ is the Fisher information, then: $\mathrm{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$
2. **Efficient Estimator**: An estimator $\hat{\theta}$ is said to be efficient if it attains the Cramér-Rao lower bound. Therefore: $\mathrm{Var}(\hat{\theta}) = \frac{1}{I(\theta)}$
3. **Assume Another Unbiased Estimator**: Let's assume there exists another unbiased estimator $\tilde{\theta}$ of θ that also achieves the CRLB. Therefore: $\mathrm{Var}(\tilde{\theta}) = \frac{1}{I(\theta)}$
4. **Asymptotic Distribution**: For large sample sizes, both estimators $\hat{\theta}$ and $\tilde{\theta}$ are normally distributed with the same mean θ and variance $\frac{1}{I(\theta)}$: $\hat{\theta} \sim N\left(\theta, \frac{1}{I(\theta)}\right)$ $\tilde{\theta} \sim N\left(\theta, \frac{1}{I(\theta)}\right)$
5. **Equivalence**: If two estimators are asymptotically normally distributed with the same mean and variance, they are asymptotically equivalent. This means that for large sample sizes, the difference between the estimators converges to zero in probability: $\hat{\theta} - \tilde{\theta} \xrightarrow{p} 0$
6. **Conclusion**: Since $\hat{\theta}$ and $\tilde{\theta}$ are asymptotically equivalent and achieve the same variance (CRLB), they must be essentially the same estimator. Thus, the efficient estimator $\hat{\theta}$ is unique among the class of unbiased estimators that achieve the CRLB.